



ALLIANCE
UNIVERSITY

NAAC
GRADE A+
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ER DIAGRAM DESIGN AND RELATIONAL SCHEMA

Assignment 2

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OBJECTIVE

To design a complete Entity Relationship (ER) Diagram for the Food Delivery Management System and convert it into a Relational Schema by identifying entities, attributes, primary keys, foreign keys, relationships, and cardinalities.

ENTITIES AND ATTRIBUTES

Entity 1: CUSTOMER

Column	Key
Customer_ID	Primary Key
Customer_Name	-
Phone	-
Email	-
Address	-

Entity 2: RESTAURANT

Column	Key
Restaurant_ID	Primary Key
Restaurant_Name	-
Phone	-
Rating	-
Address	-

Entity 3: MENU_ITEM

Column	Key
Item_ID	Primary Key
Restaurant_ID	Foreign Key
Item_Name	-
Category	-
Price	-
Availability	-

Entity 4: ORDERS

Column	Key
Order_ID	Primary Key
Customer_ID	Foreign Key
Restaurant_ID	Foreign Key
Order_Date	-
Total_Amount	-
Order_Status	-

Entity 5: ORDER_ITEM

Column	Key
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Item_ID	Foreign Key
Quantity	-
Subtotal	-

Entity 6: PAYMENT

Column	Key
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Date	-
Payment_Method	-
Amount	-
Payment_Status	-

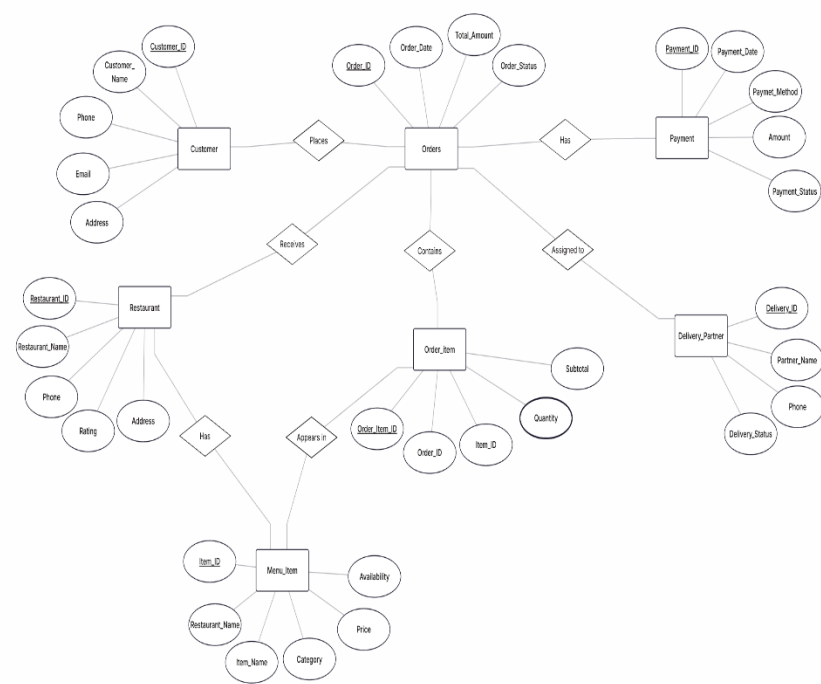
Entity 7: DELIVERY_PARTNER

Column	Key
Delivery_ID	Primary Key
Order_ID	Foreign Key
Partner_Name	-
Phone	-
Delivery_Status	-

Relationships and Cardinalities

Entity 1	Relationship	Entity 2	Cardinality
CUSTOMER	Places	ORDERS	1:M
RESTAURANT	Has	MENU_ITEM	1:M
RESTAURANT	Receives	ORDERS	1:M
ORDERS	Contains	ORDER_ITEM	1:M
MENU_ITEM	Appears in	ORDER_ITEM	1:M
ORDERS	Has	PAYMENT	1:1
ORDERS	Assigned to	DELIVERY_PARTNER	1:1

ER DIAGRAM



RELATIONAL SCHEMA

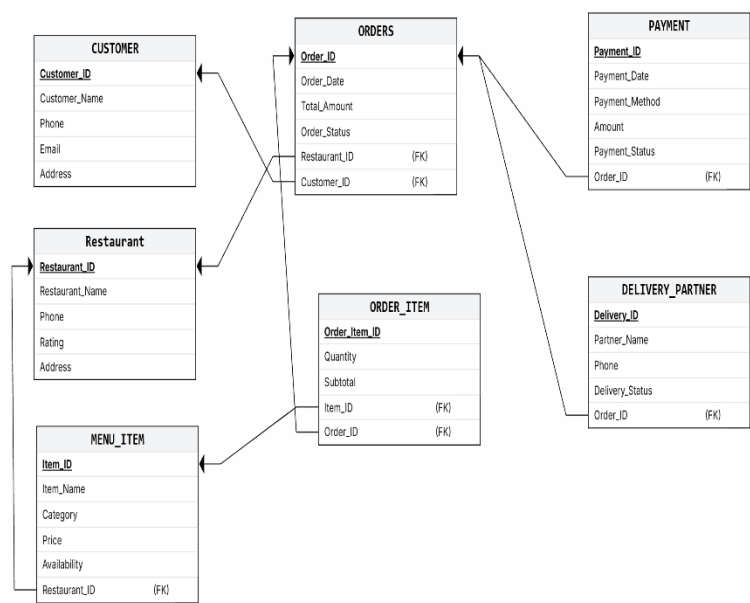


TABLE	PRIMARY KEY	FOREIGN KEY
CUSTOMER	Customer_ID	-
RESTAURANT	Restaurant_ID	-
MENU_ITEM	Item_ID	Restaurant_ID
ORDERS	Order_ID	Customer_ID, Restaurant_ID
ORDER_ITEM	Order_Item_ID	Order_ID, Item_ID
PAYMENT	Payment_ID	Order_ID
DELIVERY_PARTNER	Delivery_ID	Order_ID

RELATIONSHIP EXPLANATION

Customer – Orders:

One customer can place many orders. Therefore, the relationship is **1:M**.

Restaurant – Menu Item:

One restaurant can have many menu items. Therefore, the relationship is **1:M**.

Restaurant – Orders:

One restaurant can receive many orders. Therefore, the relationship is **1:M**.

Orders – Order Item:

One order can contain many order items. Therefore, the relationship is **1:M**.

Menu Item – Order Item:

One menu item can appear in many order items. Therefore, the relationship is **1:M**.

Orders – Payment:

Each order has one payment record. Therefore, the relationship is **1:1**.

Orders – Delivery Partner:

Each order is assigned to one delivery record. Therefore, the relationship is **1:1**.